



ESCUELA DE INGENIERÍA  
FACULTAD DE INGENIERÍA

# Lecture 15: Profit Function

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ICS2523 Microeconomics

# Lecture 15

Last lecture, we studied

- Distinguished the Short- and Long-run
- Understood how the manager of a firm thinks to **maximise profits**
- Learnt about marginal revenue
- Estimated the supply curve in the short-run

In today's lecture, we will learn about the **Profit Function**

- Do an activity about marginal revenue
- Talk about the profit function
- Learn how to estimate the producer's surplus (in the short run)
- Learn how to estimate the producer's surplus (in the short run)
- Learn the Partial Equilibrium Competitive Model
- Do a comparative statics analysis of the long-run equilibrium

# 16. Supply Curve in the SR

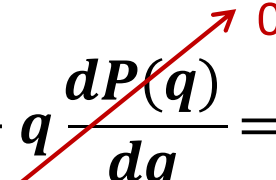
## (Review)

### Short-run Supply Curve by a Price-taking Firm:

We are going to study the **supply decision** of a price-taking firm.

The firm's short-run supply curve shows how much it will produce at various possible output prices.

To maximize its profit, the firm maximizes its revenue minus its costs::

$$MR(q) = P + q \frac{dP(q)}{dq} = MC(q)$$


If the firm is a price taker, the price  $P$  is given  $\rightarrow MC = P^*$

So, the **supply curve should be the marginal cost curve**...but should it be over its entire interval?

# 16. Supply Curve in the SR

## (Review)

### Short-run Supply Curve by a Price-taking Firm:

If the company is a price taker:

- a) When maximizing profit, the first order condition is:  $MR = MC \rightarrow P = MC$
- b) The second order condition is  $\rightarrow \partial MC / \partial q > 0$   
*Therefore, the **Marginal Cost** curve must be increasing.*
- c) When should the company close?

# 16. Supply Curve in the SR

## (Review)

### Short-run Supply Curve by a Price-taking Firm:

If the company is a price taker:

c) When should the company close?

In the short run, the firm must pay its fixed costs (even if the firm closes)

$$C_{SR} = \textit{Fixed Costs} + \textit{Variable Costs}_{SR}$$

If the firm offers  $q_2$  at  $P_2$ , the firm's profits will be:

$$\pi_{open} = P_2 q_2 - \textit{Fixed Costs} - VC_{SR}$$

If the firm closes, it will have to pay the fixed costs, and its profits will be:

$$\pi_{closed} = 0 - \textit{Fixed Costs}$$

# 16. Supply Curve in the SR

## (Review)

Short-run Supply Curve by a Price-taking Firm:

If the company is a price taker:

c) When should the company close?

To offer the good:

$$\pi_{Open} > \pi_{Closed}$$

$$Pq - \text{Fixed Costs} - VC_{SR} > -\text{Fixed Costs}$$

$$Pq > VC_{SR}$$

$$P > \frac{VC_{SR}}{q} = AVC_{SR}$$

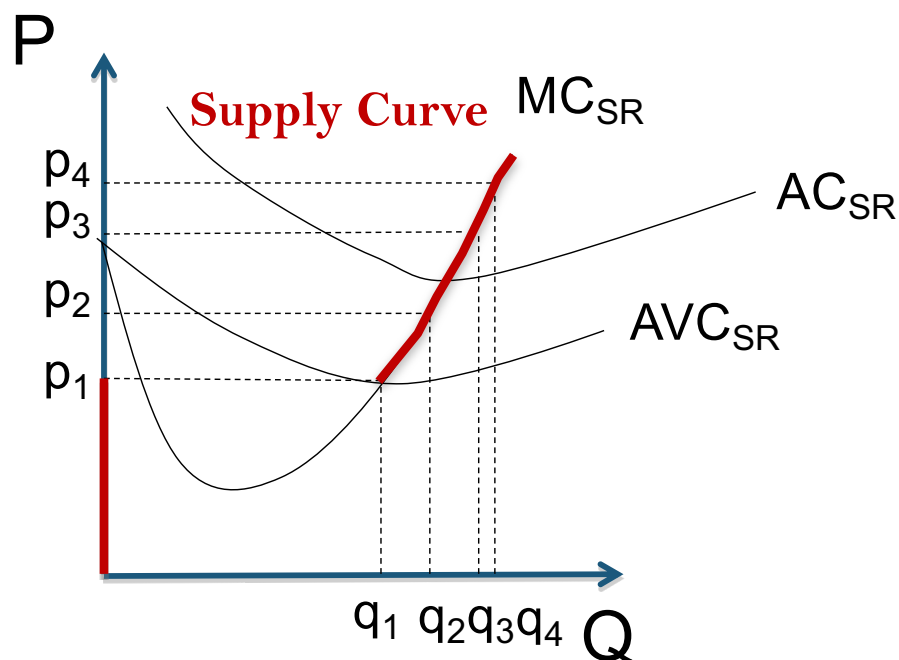
**Attention:** in the short run, the firm does not shut down when the price is less than the AC. It shuts down when the price is less than the AVC (although profits are negative the firm will prefer to supply this good, since it has to pay the fixed cost)

# 16. Supply Curve in the SR (Review)

## Short-run Supply Curve by a Price-taking Firm:

If  $P^* = P_3$ ,  $q_3$  provides maximum profits,  $\pi > 0$ .

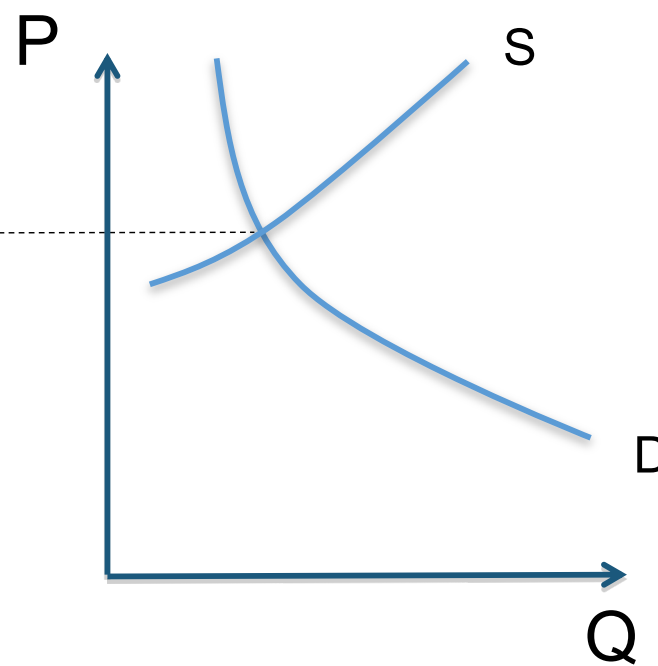
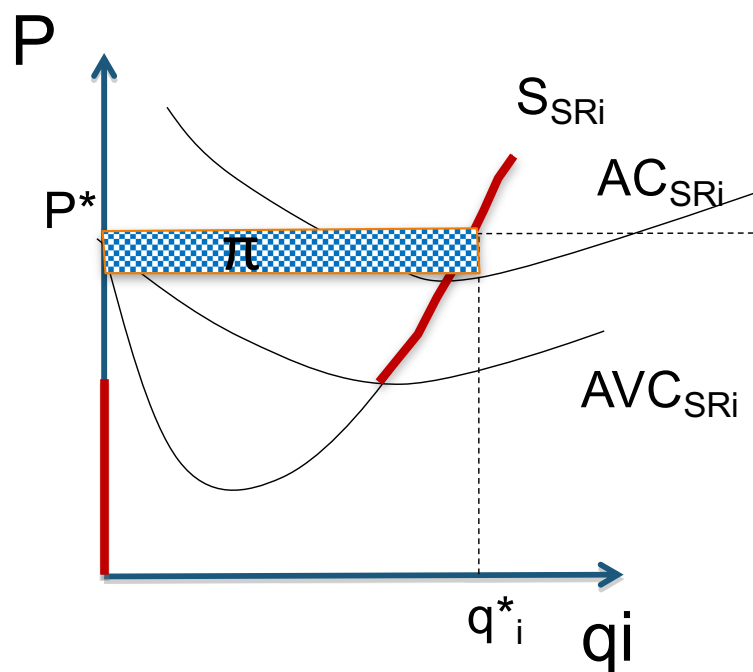
$$\text{Supply curve} = \begin{cases} MC = P & \text{for prices above } AVC_{SR} \\ 0 & \text{for prices below } AVC_{SR} \end{cases}$$



# 16. Supply Curve in the SR (Review)

Do not forget the bigger picture!

- The industry where this firm belongs has its own equilibrium where  $S=D \rightarrow$  that's how we get the price  $P^*$
- With that price we can find  $q_i^*$  and see the firm's benefit ( $\pi$ )



# Marginal Revenue

Producer Theory

# 15. Marginal Revenue

Elasticity of the demand curve is:

$$\varepsilon_{q,p} = \frac{dq / q}{dp / p} = \frac{dq}{dp} \frac{p}{q}$$

$$MR(q) = p + q \frac{dp}{dq} = p \left[ 1 + \frac{q}{p} \frac{dp}{dq} \right] = p \left[ 1 + \frac{1}{\varepsilon_{q,p}} \right]$$

# Activity 1

Imagine a firm faces a linear demand function

$$(Q = a - b P)$$

- a) Find the elasticity of the demand curve in terms of  $P$
- b) Graph the demand curve and show elasticity when  $P=0$  and when  $Q=0$
- c) Find  $P$  and  $Q$  so elasticity's absolute value is 1

$$\varepsilon_{q,p} = \frac{dq / q}{dp / p} = \frac{dq}{dp} \frac{p}{q}$$

# 15. Marginal Revenue

## Price-marginal cost markup:

When a firm maximizes profits:

$$MR(q) = MC$$

$$\Rightarrow MC = p \left[ 1 + \frac{1}{\varepsilon_{q,p}} \right]$$
$$\frac{p}{MC} = \left[ \frac{\varepsilon_{q,p}}{1 + \varepsilon_{q,p}} \right]$$

Markup of price over MC depends on the elasticity of demand that the firm faces.

Note: elasticity of demand MUST be elastic

Markup over marginal cost will be higher the closer the elasticity of demand is to -1.

# Profit Function

Producer Theory

# 17. Profit Function

Shows the firm's maximized profits as depending only on the prices that the firm faces

The **economic profits** are:

$$\pi = Pq - C = P \cdot f(k, l) - rk - wl$$

Which are the variables **under the firm's control**?

Firm chooses **level of inputs** ( $k, l$ ) in order to maximize profits and treats prices ( $P, r$ , and  $w$ ) as fixed parameters

# 17. Profit Function

→ The firm's profit function shows its maximal profits as a function of the prices that the firm faces:

$$\Pi(P, r, w) \max_{\{k, l\}} \pi(k, l) = \max_{\{k, l\}} [P \cdot f(k, l) - rk - wl]$$

# 17. Profit Function

## Properties of the Profit function

1. Homogeneous of degree 1 in all prices
  - MC are homogeneous of degree 1 in input prices
2. Profit functions are nondecreasing in output price ( $P$ )
3. Profit functions are nonincreasing in inputs prices ( $r$  and  $w$ )
4. Profit functions are convex in output prices

$$\frac{\Pi(P_1, r, w) + \Pi(P_2, r, w)}{2} \geq \Pi\left(\frac{P_1 + P_2}{2}, r, w\right)$$

# 17. Profit Function

## Properties of the Profit function

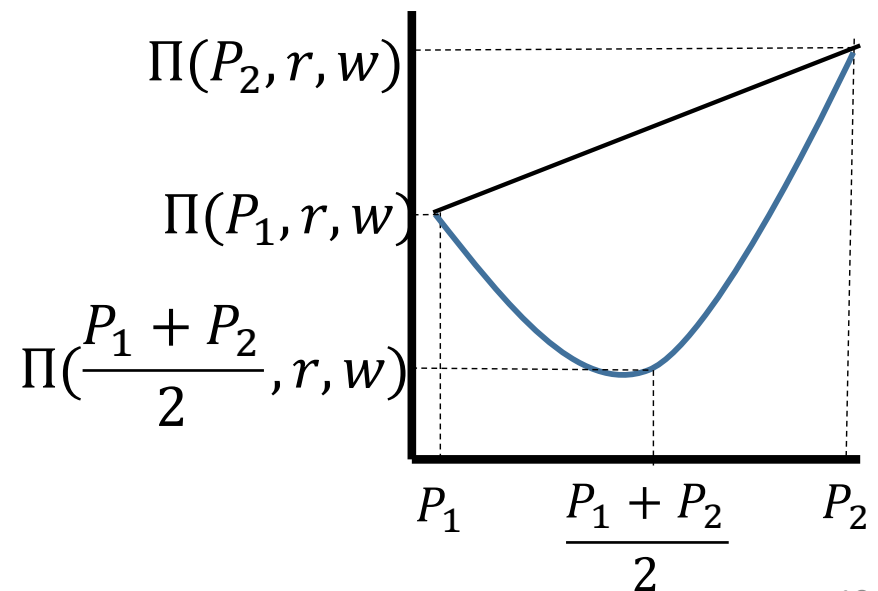
4. Profit functions are convex in output prices

$$\frac{\Pi(P_1, r, w) + \Pi(P_2, r, w)}{2} \geq \Pi\left(\frac{P_1 + P_2}{2}, r, w\right)$$

This important feature of profit functions says that the profits obtainable by averaging those available from two different output prices **will be at least as large** as those obtainable from the average of the two prices.

(Could be any linear combination of prices:

$tP_1 + (1-t)P_2$ , where  $0 < t < 1$ )



# 17. Profit Function

## Properties of the Profit function

4. Profit functions are convex in output prices

More formally, let  $P_3 = \frac{P_1 + P_2}{2}$  and let  $q_i, k_i, l_i$  represent the profit-maximizing output choices for these prices.

$$\begin{aligned}\Pi(P_3, r, w) &= P_3 q_3 - r k_3 - w l_3 \\ &= \frac{P_1 q_3 - r k_3 - w l_3}{2} + \frac{P_2 q_3 - r k_3 - w l_3}{2} \\ &\leq \frac{P_1 q_1 - r k_1 - w l_1}{2} + \frac{P_2 q_2 - r k_2 - w l_2}{2} \\ &\equiv \frac{\Pi(P_1, r, w) + \Pi(P_2, r, w)}{2}\end{aligned}$$

# Activity 2

Since the Profit Function is homogeneous of degree 1 in prices, what happens with the quantity produced when there is inflation?

- a) Increases
- b) Stays constant
- c) Decreases



Wooclap code: **QPCEPM**

<https://app.wooclap.com/QPCEPM?from=even-t-page>

# 17. Profit Function

## Envelope Theorem

How the optimal value for a particular function changes when a parameter of the function changes

In the case of the profit function:

$$\frac{\partial \Pi(P, r, w)}{\partial P} = q(P, r, w)$$

$$\frac{\partial \Pi(P, r, w)}{\partial r} = -k(P, r, w)$$

$$\frac{\partial \Pi(P, r, w)}{\partial w} = -l(P, r, w)$$

# Producer Surplus

Producer Theory

# 18. Producer Surplus (short-run)

As the profit function is nondecreasing in output prices, if  $P_2 > P_1$  then:

$$\Pi(P_2, \dots) \geq \Pi(P_1, \dots)$$

We can measure the welfare gain to the firm from the price change as:

$$\text{Welfare Gain: } \Pi(P_2, \dots) - \Pi(P_1, \dots)$$

# Activity 3

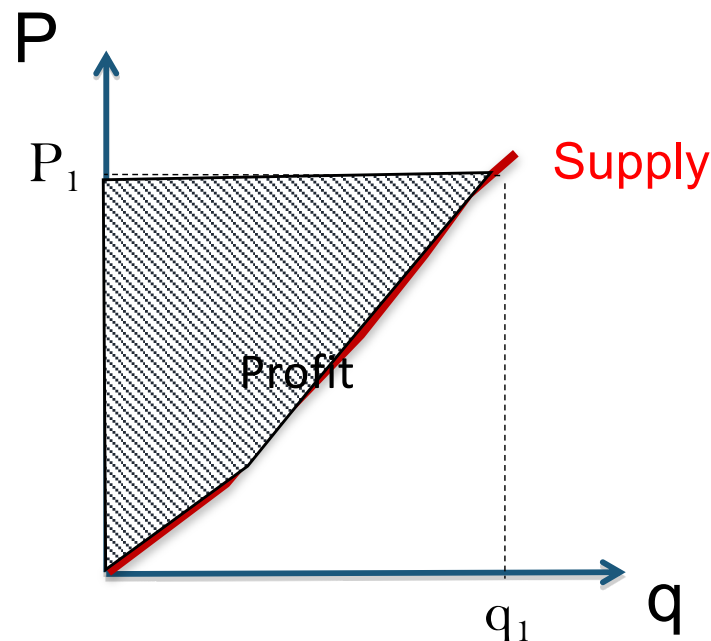
Draw a graph of the supply curve of a firm and find a way to show graphically the welfare change when the price of the output increases.

# 18. Producer Surplus (short-run)

Producer surplus =

$$\Pi(P_1, \dots) - \Pi(P_0, \dots) = \int_{P_0}^{P_1} q(P) dP$$

Is the **extra return that producers earn** by making transactions at the market price over and above what they would earn if nothing were produced



# 18. Producer Surplus (short-run)

Remember that we are studying the **short-run**

What happens with  $\Pi$  at the **shutdown price**? How does this affect the producer surplus in the SR?

At the shutdown price ( $P_0$ ), the firm produces no output, thus profits at shutdown price are solely made up of losses of all fixed costs  $\rightarrow \Pi(P_0, r, w) = -rk$

$$\text{Producer Surplus} = \Pi(P_1, r, w) - \Pi(P_0, r, w)$$

$$\text{Producer Surplus} = \Pi(P_1, r, w) + rk$$

**Fixed Costs**

Producer surplus is given by current profits being earned plus short-run fixed costs

# 18. Producer Surplus (long-run)

Producer Surplus<sub>LR</sub> =

$$\begin{aligned}\Pi(P_1) &= P_1 \cdot q_1 - C_{LR}(q_1) \\ &= P_1 \cdot q_1 - AC_{LR}(q_1) \cdot q_1 \\ &= (P_1 - AC_{LR}(q_1)) \cdot q_1 \\ &= P_1 \cdot q_1 - \int_0^{q_1} MC_{LR}(q) dq\end{aligned}$$

# Competitive Markets

Fourth Chapter

# Competitive Markets

So far, we have developed models to explain the **demand for goods** (utility-maximising individuals) and the **supply for goods** (profit-maximising firms).

Now, we are going to bring these models together to **describe the process by which prices are determined**.

We will focus on only one specific model of price determination  
→ **Perfect competitive model**.

The perfect competitive model assumes a large enough number of demanders and suppliers of each good so that each must be a price taker.

# Competitive Markets

First, we will analyse the **partial equilibrium model of price determination**. This model illustrates a partial equilibrium view of price determination because it focuses on only a single market.

Since the partial equilibrium model is inappropriate for examining relationships among markets, later on, we will study the **general equilibrium models**: we will see how an entire economy can be viewed as a system of interconnected competitive markets that determine all prices simultaneously. We will also examine how the welfare consequences of various economic questions can be studied with this model.

# The Partial Equilibrium Competitive Model

Competitive Markets

# 1. The Partial Equilibrium Competitive Model

What is a **perfectly competitive industry**?

1. There is a large number of firms, each producing the same homogeneous product.
2. Each firm attempts to maximise profits
3. Each consumer attempts to maximise his utility
4. Each firm is a price-taker
5. Prices are assumed to be known by all market participants (information is perfect)
6. Transactions are costless
7. Free entry and exit of firms (only for the long-run)

# 1. The Partial Equilibrium Competitive Model

<http://www.youtube.com/watch?v=9Hxy-TuX9fs>

Remember individual's demand? (Marshallian)

quantity of  $x$  demanded =  $x(P_x, P_y, I)$

$$\text{Market demand for } X = \sum_{i=1}^n x_i(P_x, P_y, I_i)$$

How is this curve (market demand) done graphically?

→ HORIZONTAL SUM

# Timing of the Supply Response and Pricing

Competitive Markets

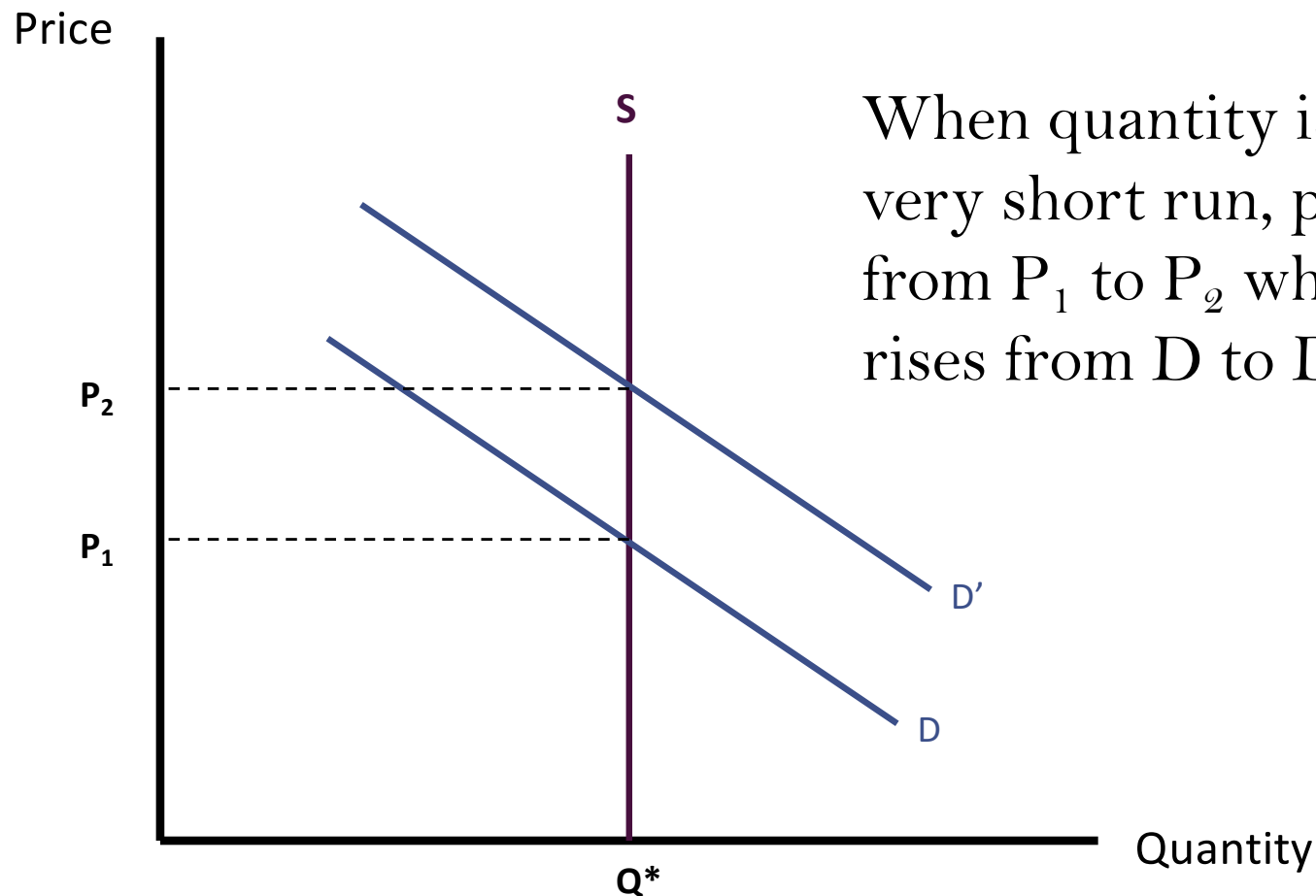
## 2. Timing of Supply Response

In the competitive pricing analysis, it is important to decide the length of time to be allowed for a supply response to changing demand conditions.

- Pricing in the **very short-run** → no supply response  
→ Supply is fixed at  $Q^*$
- Pricing in the **short-run** → number of firms in an industry is fixed (they can adjust the quantity they are producing)
- Pricing in the **long-run** → new firms may enter an industry

# 3. Pricing in the very short-run

## Pricing in the very short-run



# 3. Pricing in the short-run

## Short-run market supply curve

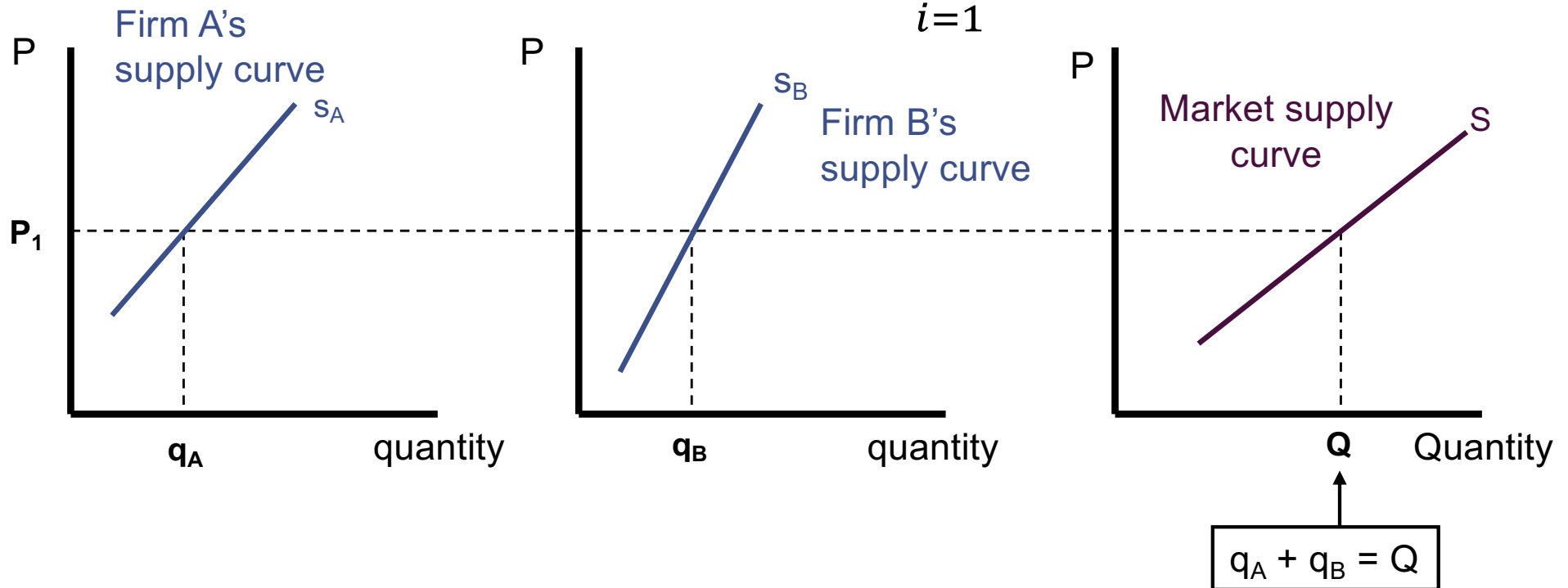
- The quantity of output supplied to the entire market in the short-run is the sum of the quantities supplied by each firm.
- This quantity depends on the price.
- Since each firm's supply curve has a positive slope, the market supply curve will also have a positive slope (Why?)
- Short-run market supply function:

$$Q_S(P, r, w) = \sum_{i=1}^n q_i(P, r, w)$$

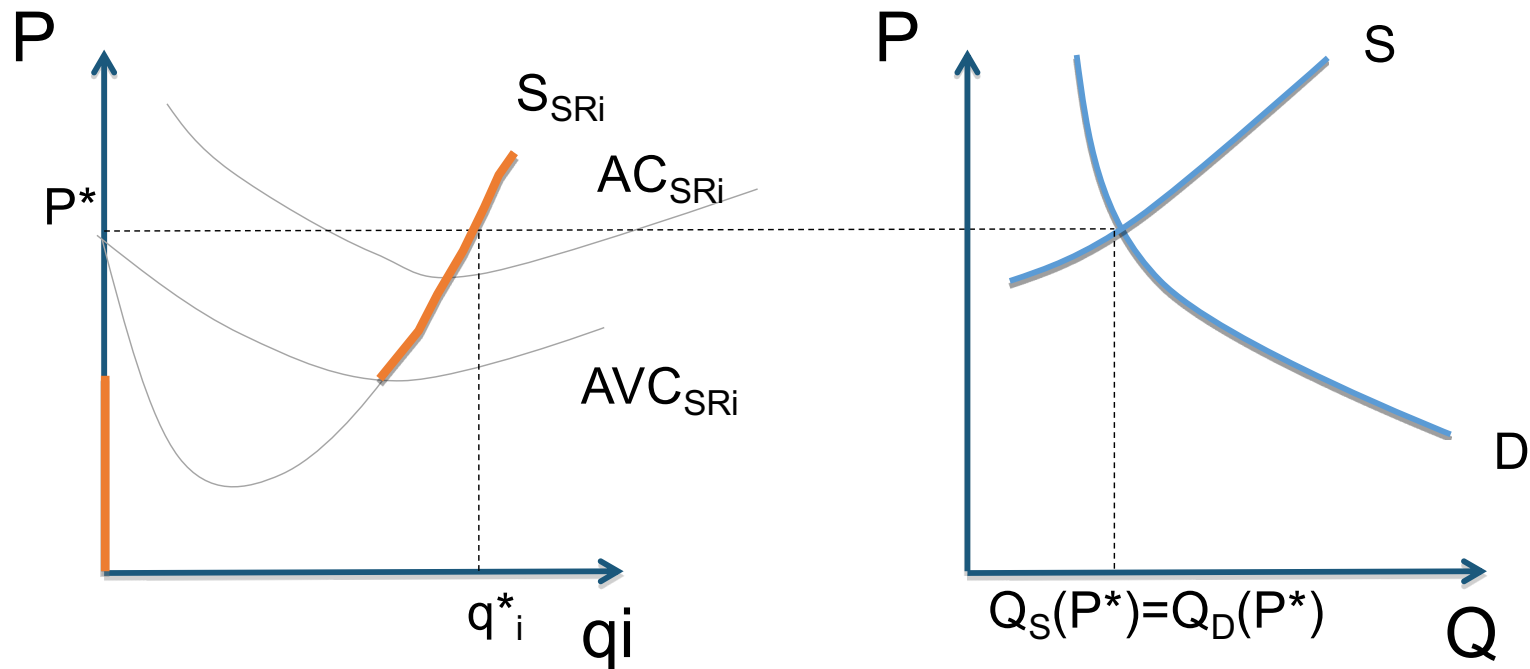
# 3. Pricing in the short-run

## Short-run market supply curve

$$Q_S(P, r, w) = \sum_{i=1}^n q_i(P, k, w)$$



### 3. Pricing in the short-run



An equilibrium price ( $P^*$ ) is one at which the quantity demanded is equal to the quantity supplied  $\rightarrow$  at this price no one has incentives to alter their economic decisions

$$Q_D(P^*, P', I) = Q_S(P^*, r, w)$$
$$Q_D(P^*) = Q_S(P^*)$$

# 3. Pricing in the short-run

Short-run supply elasticity

$$e_{S,P} = \frac{\partial Q_S}{\partial P} \cdot \frac{P}{Q_S}$$

Please note

- 1) Supply elasticity is POSITIVE → Why?
- 2) Relationship between elasticity and MC:
  - Low values of elasticity → It takes relatively large changes in price to induce firms to change their output levels → MC increase rapidly
  - High values of elasticity → Small increases in market price lead to a relatively large supply response by firms → MC do not rise steeply

# 3. Pricing in the short-run

## Shifts in the supply curve (short-run)

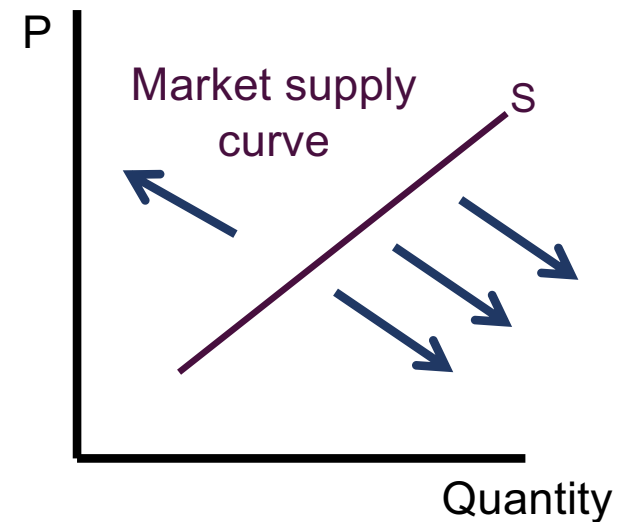
What happens to the supply curve in the short-run if...

The number of fixed inputs increases?

The price of the variable inputs increase?

There is a technological change?

Number of producers increase?



# Activity 4

Which of the following changes will shift the SR supply curve for leather shoes?

- a) Increase in the rent of the shoe factory, which is fixed for 2 years
- b) Decrease in the price of leather, which is bought depending on the level of production
- c) Increase in the price of leather shoes
- d) Decrease in the price of boots



Wooclap code: **QPCEPM**

<https://app.wooclap.com/QPCEPM?from=even-t-page>

# Individual Activity

In a piece of paper write down **2-3 concepts** that you learnt today. Please write down the concept and what you learnt about it, in your own words.

If you have any **questions** about we discussed today, write it down too. We will go through them at the beginning of next class.

Finally, if there is a word you didn't understand, write it down so it can be part of the **glossary**.